

Задание №1

Создать приложение по образцу

Листинг:

```
using MySql.Data.MySqlClient;
using System;
using System.Collections.Generic;
using System.ComponentModel;
using System.Data;
using System.Drawing;
using System.Linq;
using System.Text;
using System.Threading.Tasks;
using System.Windows.Forms;
using System.Windows.Forms.DataVisualization.Charting;

namespace WindowsFormsApp2
{
    public partial class Form1 : Form
    {
        MySqlConnection connection;
        public Form1() => InitializeComponent();
        private void Form1_Load(object sender, EventArgs e)
        {
            try
            {
                connection = new
                MySqlConnection("datasource=localhost;port=3306;username=root;password=root;SslMode=none");
                connection.Open();
                MySqlDataAdapter adapter = new MySqlDataAdapter("SELECT *
                FROM Квалификация.Преподаватели", connection);
                DataSet ds = new DataSet();
                adapter.Fill(ds, "Преподаватели");
                dataGridView3.DataSource = ds.Tables["Преподаватели"];
                for (int i = 0; i < dataGridView3.ColumnCount; i++)
                {
                    comboBox4.Items.Add(dataGridView3.Columns[i].Name);
                }
                connection.Close();

                connection.Open();
                MySqlDataAdapter adapter2 = new MySqlDataAdapter("SELECT *
                FROM Квалификация.Часы_занятий", connection);
                DataSet ds2 = new DataSet();
                adapter2.Fill(ds2, "Часы_занятий");
```

```
dataGridView4.DataSource = ds2.Tables["Часы_занятий"];
for (int i = 0; i < dataGridView4.ColumnCount; i++)
{
    comboBox5.Items.Add(dataGridView4.Columns[i].Name);
}
connection.Close();

connection.Open();
using (MySqlDataReader reader = GetCommand("SELECT
TABLE_NAME FROM INFORMATION_SCHEMA.TABLES WHERE
TABLE_TYPE = 'BASE TABLE' AND TABLE_SCHEMA =
'Квалификация';").ExecuteReader())
{
    while (reader.Read())
    {
        comboBox2.Items.Add(reader.GetString(0));
        comboBox3.Items.Add(reader.GetString(0));
        comboBox1.Items.Add(reader.GetString(0));
    }
}
connection.Close();
}
catch (Exception ex) { MessageBox.Show(ex.Message); }
}

private MySqlCommand GetCommand(string sql) => new
MySqlCommand(sql, connection);

private void button5_Click(object sender, EventArgs e)
{
    try
    {
        connection.Open();
        MySqlDataAdapter adapter2 = new MySqlDataAdapter("SELECT " +
comboBox4.Text + ", " + comboBox5.Text + " FROM
Квалификация.Часы_занятий, Квалификация.Преподаватели WHERE
Преподаватели.Код = Часы_занятий.Код_преподавателя;", connection);
        DataSet ds = new DataSet();
        adapter2.Fill(ds);
        dataGridView5.DataSource = ds.Tables[0];
    }
    catch (Exception ex) { MessageBox.Show(ex.Message); }
    connection.Close();
}

private void button6_Click(object sender, EventArgs e)
```

```
{
    connection.Open();
    MySqlDataAdapter adapter = new MySqlDataAdapter("SELECT * FROM
Квалификация.Часы_занятий", connection);
    DataSet ds = new DataSet();
    adapter.Fill(ds, "Часы_занятий");
    dataGridView6.DataSource = ds.Tables["Часы_занятий"];
    connection.Close();

    connection.Open();
    string id_query = $"select Код_группы FROM
Квалификация.Часы_занятий";
    string new_tablecol_query = $"select Количество_часов FROM
Квалификация.Часы_занятий";

    using (MySqlDataReader reader =
GetCommand(new_tablecol_query).ExecuteReader())
    {
        int i = 0;
        while (reader.Read())
        {
            chart1.Series.Add(new Series());
            chart1.Series[i].Points.AddY(reader.GetString(0));
            i++;
        }
    }

    using (MySqlDataReader reader2 =
GetCommand(id_query).ExecuteReader())
    {
        int i = 0;
        while (reader2.Read())
        {
            chart1.Series[i].AxisLabel = reader2.GetString(0);
            i++;
        }
        chart1.AlignDataPointsByAxisLabel();
    }
    connection.Close();
}

private void button4_Click(object sender, EventArgs e)
{
    try
    {
```

```
connection.Open();
MySQLDataAdapter adapter = new MySQLDataAdapter("SELECT *
FROM Квалификация." + comboBox2.Text, connection);
DataSet ds = new DataSet();
adapter.Fill(ds, comboBox2.Text);
dataGridView1.DataSource = ds.Tables[comboBox2.Text];
connection.Close();
if (checkBox1.Checked)
{
    connection.Open();
    MySQLDataAdapter adapter2 = new MySQLDataAdapter("SELECT *
FROM Квалификация." + comboBox3.Text, connection);
    DataSet ds2 = new DataSet();
    adapter2.Fill(ds, comboBox3.Text);
    dataGridView2.DataSource = ds.Tables[comboBox3.Text];
    connection.Close();
}
}
catch (Exception ex) { MessageBox.Show(ex.Message); }
}
// ----- CLOSED -----
```

```
private void comboBox1_SelectedIndexChanged(object sender, EventArgs e)
{
    List<Label> lv = new List<Label>();
    lv.Add(label4);
    lv.Add(label5);
    lv.Add(label6);
    lv.Add(label7);
    lv.Add(label8);
    lv.Add(label9);

    label4.Text = "";
    label5.Text = "";
    label6.Text = "";
    label7.Text = "";
    label8.Text = "";
    label9.Text = "";
    textBox1.Enabled = true;
    textBox2.Enabled = true;
    textBox3.Enabled = true;
    textBox4.Enabled = true;
    textBox5.Enabled = true;
```

```
textBox6.Enabled = true;

if (comboBox1.Text == "специальности")
{
    textBox2.Enabled = false;
    textBox3.Enabled = false;
    textBox4.Enabled = false;
    textBox5.Enabled = false;
    textBox6.Enabled = false;
    connection.Open();
    using (MySqlDataReader adapter2 = GetCommand("SELECT
COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE
TABLE_NAME = 'Специальности';").ExecuteReader())
    {
        List<string> vs = new List<string>();
        while (adapter2.Read())
        {
            vs.Add(adapter2.GetString(0));
        }
        connection.Close();
        for (int i = 0; i < 4; i++)
        {
            if (i + 1 < vs.Count)
                lv[i].Text = vs[i + 1];
        }
    }
}
if (comboBox1.Text == "отделения")
{
    textBox3.Enabled = false;
    textBox4.Enabled = false;
    textBox5.Enabled = false;
    textBox6.Enabled = false;
    connection.Open();
    using (MySqlDataReader adapter2 = GetCommand("SELECT
COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE
TABLE_NAME = 'Отделения';").ExecuteReader())
    {
        List<string> vs = new List<string>();
        while (adapter2.Read())
        {
            vs.Add(adapter2.GetString(0));
        }
        connection.Close();
        for (int i = 0; i < 4; i++)
```

```
        {
            if (i + 1 < vs.Count)
                lv[i].Text = vs[i + 1];
        }
    }
}
if (comboBox1.Text == "группы")
{
    textBox2.Enabled = false;
    textBox3.Enabled = false;
    textBox4.Enabled = false;
    textBox5.Enabled = false;
    textBox6.Enabled = false;
    connection.Open();
    using (MySqlDataReader adapter2 = GetCommand("SELECT
COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE
TABLE_NAME = 'Группы;").ExecuteReader())
    {
        List<string> vs = new List<string>();
        while (adapter2.Read())
        {
            vs.Add(adapter2.GetString(0));
        }
        connection.Close();
        for (int i = 0; i < 4; i++)
        {
            if (i + 1 < vs.Count)
                lv[i].Text = vs[i + 1];
        }
    }
}
if (comboBox1.Text == "слушатели")
{
    textBox5.Enabled = false;
    textBox6.Enabled = false;
    label9.Text = "";
    connection.Open();
    using (MySqlDataReader adapter2 = GetCommand("SELECT
COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE
TABLE_NAME = 'Слушатели;").ExecuteReader())
    {
        List<string> vs = new List<string>();
        while (adapter2.Read())
        {
            vs.Add(adapter2.GetString(0));
```

```
    }
    connection.Close();
    for (int i = 0; i < 4; i++)
    {
        if (i + 1 < vs.Count)
            lv[i].Text = vs[i + 1];
    }
}
if (comboBox1.Text == "преподаватели")
{
    textBox6.Enabled = false;
    connection.Open();
    using (MySqlDataReader adapter2 = GetCommand("SELECT
COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE
TABLE_NAME = 'Преподаватели;').ExecuteReader())
    {
        List<string> vs = new List<string>();
        while (adapter2.Read())
        {
            vs.Add(adapter2.GetString(0));
        }
        connection.Close();
        for (int i = 0; i < 5; i++)
        {
            if (i + 1 < vs.Count)
                lv[i].Text = vs[i + 1];
        }
    }
}
if (comboBox1.Text == "занятия")
{
    textBox4.Enabled = false;
    textBox5.Enabled = false;
    textBox6.Enabled = false;
    connection.Open();
    using (MySqlDataReader adapter2 = GetCommand("SELECT
COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE
TABLE_NAME = 'Занятия;').ExecuteReader())
    {
        List<string> vs = new List<string>();
        while (adapter2.Read())
        {
            vs.Add(adapter2.GetString(0));
        }
    }
}
```

```
        connection.Close();
        for (int i = 0; i < 4; i++)
        {
            if (i + 1 < vs.Count)
                lv[i].Text = vs[i + 1];
        }
    }
}
if (comboBox1.Text == "часы_занятий")
{
    textBox5.Enabled = false;
    textBox6.Enabled = false;
    connection.Open();
    using (MySqlDataReader adapter2 = GetCommand("SELECT
COLUMN_NAME FROM INFORMATION_SCHEMA.COLUMNS WHERE
TABLE_NAME = 'Часы_занятий';").ExecuteReader())
    {
        List<string> vs = new List<string>();
        while (adapter2.Read())
        {
            vs.Add(adapter2.GetString(0));
        }
        connection.Close();
        for (int i = 0; i < 4; i++)
        {
            if (i + 1 < vs.Count)
                lv[i].Text = vs[i + 1];
        }
    }
}

private void button1_Click(object sender, EventArgs e)
{
    if (comboBox1.Text == "специальности")
    {
        connection.Open();
        string qry = "INSERT INTO специальности (" + label5.Text + ")
VALUES (" + textBox2.Text + ");";
        using (MySqlCommand cmd = new MySqlCommand(qry, connection))
        {
            cmd.ExecuteNonQuery();
        }
        connection.Close();
    }
}
```

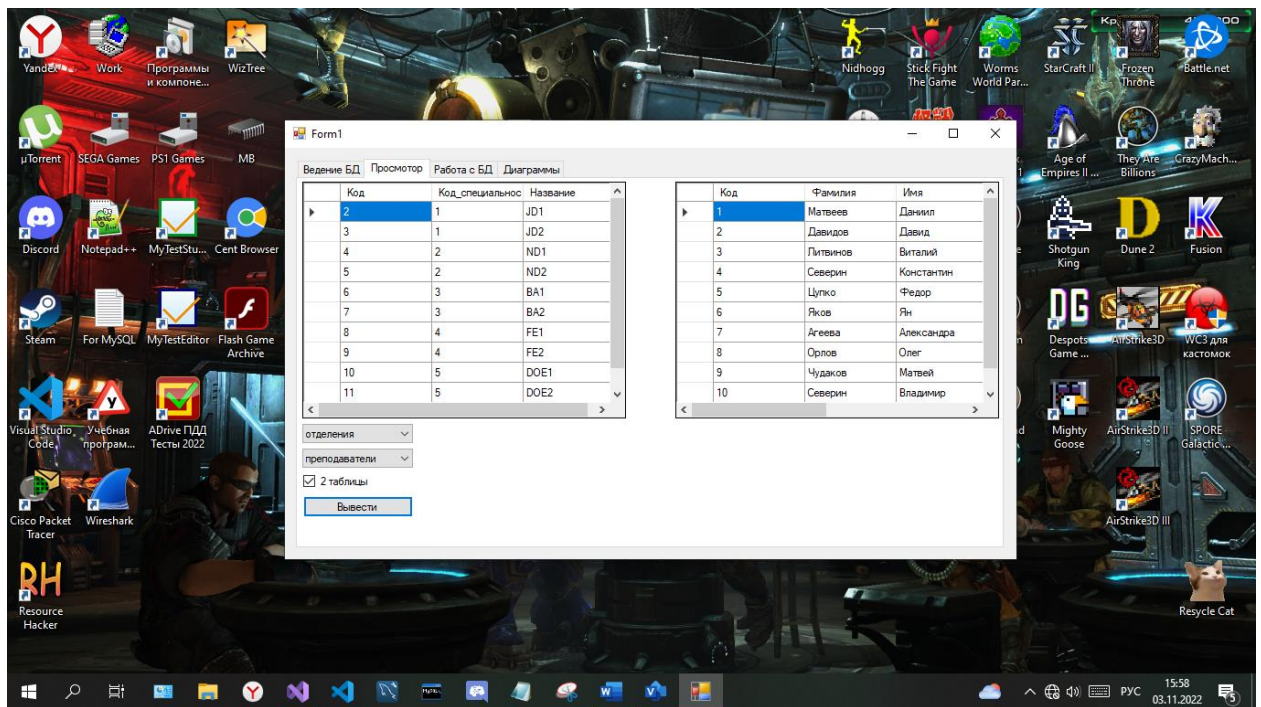
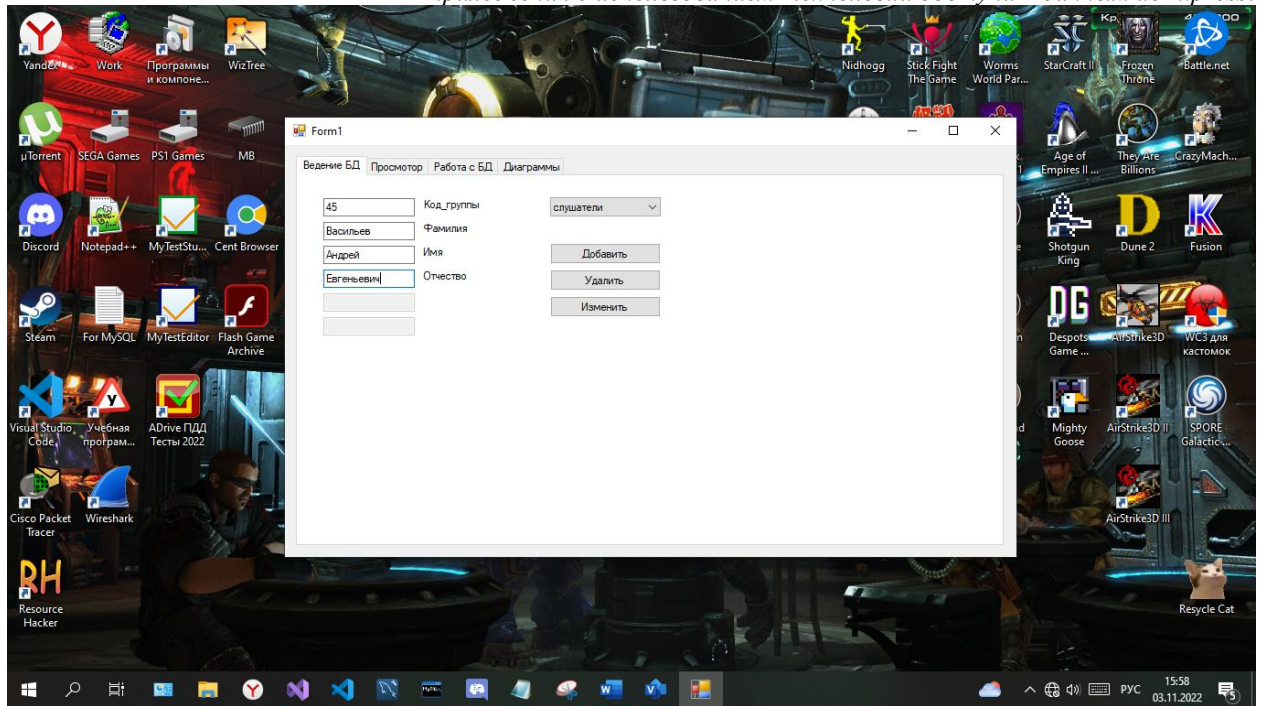


```
if (comboBox1.Text == "отделения")
{
    connection.Open();
    string qry = "INSERT INTO отделения (" + label5.Text + "," +
label6.Text + ") VALUES (" + textBox2.Text + "," + textBox3.Text + ");";
    using (MySqlCommand cmd = new MySqlCommand(qry, connection))
    {
        cmd.ExecuteNonQuery();
    }
    connection.Close();
}
if (comboBox1.Text == "группы")
{
    connection.Open();
    string qry = "INSERT INTO группы (" + label5.Text + ") VALUES (" +
textBox2.Text + ");";
    using (MySqlCommand cmd = new MySqlCommand(qry, connection))
    {
        cmd.ExecuteNonQuery();
    }
    connection.Close();
}
if (comboBox1.Text == "слушатели")
{
    connection.Open();
    string qry = "INSERT INTO Слушатели (" + label5.Text + "," +
label6.Text + "," + label7.Text + "," + label8.Text + ") VALUES (" +
textBox2.Text + "," + textBox3.Text + "," + textBox4.Text + "," + textBox5.Text +
");";
    using (MySqlCommand cmd = new MySqlCommand(qry, connection))
    {
        cmd.ExecuteNonQuery();
    }
    connection.Close();
}
if (comboBox1.Text == "преподаватели")
{
    connection.Open();
    string qry = "INSERT INTO преподаватели (" + label5.Text + "," +
label6.Text + "," + label7.Text + "," + label8.Text + "," + label9.Text + ")
VALUES (" + textBox2.Text + "," + textBox3.Text + "," + textBox4.Text + "," +
textBox5.Text + textBox6.Text + ");";
    using (MySqlCommand cmd = new MySqlCommand(qry, connection))
    {
        cmd.ExecuteNonQuery();
    }
}
```

```
    }
    connection.Close();
}
if (comboBox1.Text == "часы_занятий")
{
    connection.Open();
    string qry = "INSERT INTO часы_занятий (" + label5.Text + "," +
label6.Text + "," + label7.Text + "," + label8.Text + ") VALUES (" +
textBox2.Text + "," + textBox3.Text + "," + textBox4.Text + "," + textBox5.Text +
");";
    using (MySqlCommand cmd = new MySqlCommand(qry, connection))
    {
        cmd.ExecuteNonQuery();
    }
    connection.Close();
}
if (comboBox1.Text == "занятия")
{
    connection.Open();
    string qry = "INSERT INTO занятия (" + label5.Text + "," + label6.Text
+ "," + label7.Text + ") VALUES (" + textBox2.Text + "," + textBox3.Text + "," +
textBox4.Text + ");";
    using (MySqlCommand cmd = new MySqlCommand(qry, connection))
    {
        cmd.ExecuteNonQuery();
    }
    connection.Close();
}
}
}
}
```

Результат:

Лабораторная работа №28,29. Создание составных отчетов, отчетов с диаграммами. Разработка приложения с использованием технологии доступа к данным dbExpress.



Лабораторная работа №28,29. Создание составных отчетов, отчетов с диаграммами. Разработка приложения с использованием технологии доступа к данным dbExpress.

Form1

Ведение БД | Просмотр | Работа с БД | Диаграммы

Преподаватели

Код	Фамилия	Имя	Отчество
1	Матвеев	Даниил	М
2	Давидов	Давид	М
3	Литвинов	Виталий	О
4	Северин	Константин	М
5	Цулко	Федор	И

Часы занятий

Код	Код_преподавателя	Код_группы	Количество_часов
1	1	15	1
2	1	15	2
3	2	13	4
4	2	15	5
5	4	14	3

Отчество: [] Количество_часов: []

Вывести результат

Результирующая таблица

Отчество	Количество_часов
Матвеевич	12
Матвеевич	10
Маркович	16
Маркович	10
Маркович	18
Маркович	12

Form1

Ведение БД | Просмотр | Работа с БД | Диаграммы

Преподаватели

Код	Код_преподавателя	Код_группы	Код_группы
1	1	15	1
2	1	15	2
3	2	13	4
4	2	15	5
5	4	14	3
6	4	14	3
7	6	17	6
8	8	14	3
9	4	15	4
10	5	13	6

Сгенерировать диаграмму

Series1
Series2
Series3
Series4
Series5
Series6
Series7
Series8
Series9
Series10